Mines Paris PSL - Etilab

Pooling Contracts between Venture Capital Institutions

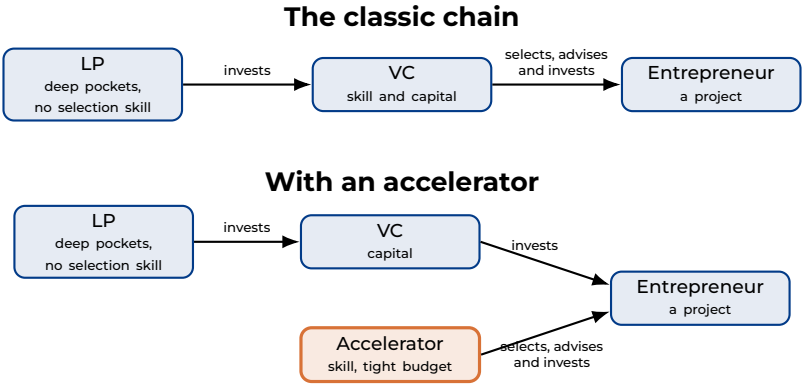
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Joint with Pierre Fleckinger

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Outline

1. Introduction
2. Literature Review
3. Contribution
4. Model
5. Policy
6. Takeaways

Who does what in venture capital

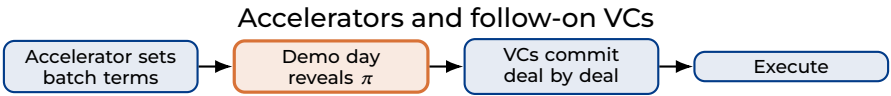


Contracts between VC institutions

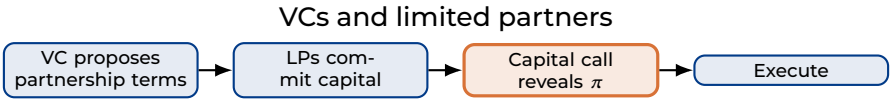
- VCs and limited partners
  - LPs commit before seeing any target company, in 93% of deals *(Lerner et al., 2022)*. Commitment is ex ante.
  - The VC invests 1–5% of the total cost, the same for every project.
  - The VC receives 20% of the return plus 2% of committed capital, the same for every project.
- Accelerators and follow-on VCs
  - A demo day reveals the batch to follow-on VCs before they invest. Commitment is ex post.
  - Terms are posted in advance and identical within a batch:

	Fixed amount, fixed share	Contingent tranche
Y Combinator	$125K for 7%	$375K uncapped MFN safe, 2022–
Techstars	$20K for 6%, 5% since 2025	$100K note capped at $3M, then MFN
500 Global	$150K for 6%	—
Entrepreneur First	$125K for 8%, London	$125K uncapped MFN safe
Antler	£125K for 8.5%, UK	£85K convertible note

Two relationships, two timings



Investors learn π before they commit: ex-post commitment.



Investors commit before they learn π: ex-ante commitment.

Literature Review

- Performance of contracts between VCs and LPs
  - Lerner and Nanda (2020), Metrick and Yasuda (2010): management fees push funds to grow at the expense of performance.
  - Axelson et al. (2009), Fang et al. (2015): blind pool contracts reduce VCs' incentives to overstate project values.
  - Gourier et al. (2024), Lerner and Nanda (2020): LPs' demand for VC is high and concentrated.
- Emergence and performance of the accelerator model
  - Ewens, Nanda and Rhodes-Kropf (2018): cheaper experiments move early-stage investors to smaller initial stakes in more startups.
  - Gonzalez-Uribe and Leatherbee (2017): accelerators increase VC fundraising and scale.
  - Kim and Wagman (2014): an accelerator picks a smaller portfolio than the efficient one and may disclose only its good signals.
  - Zarei, Rasti-Barzoki and Moon (2022): screening is the accelerator's service that matters most.
- Theory of contracts
  - Myerson (1983): an informed principal can propose one mechanism whatever the projects it holds.
  - Maskin and Tirole (1992), Holmstrom and Myerson (1983), Guesnerie and Laffont (1984), Kamien and Schwartz (1991).

This paper

We study contracts between venture capital institutions in an informed-principal model with competitive investors.

Results

- Under both regimes, pooling contracts are strategically optimal for the agency.
- Under ex-post commitment:
  - the unique optimal contract is pooling,
  - the agency's budget is always saturated,
  - some good projects always remain unfunded.

The agents

Entrepreneurs

A project needs a sunk cost $I \in (0,1)$, yields an outcome in $\{0,1\}$ with success probability π. Density f, c.d.f. F. No effort cost: the entrepreneur needs no equity.

The agency

- Budget B_a
- Knows π for each project
- Profit-maximizing

Investors

- Budget B_i
- Know only the distribution of π
- Competitive (price-taking)

Allocation $\mu = (\sigma, \rho)$: the agency contributes $\sigma(\pi)$ to the cost I and keeps a share $\rho(\pi)$ of the outcome. The investor pays the rest of the cost and receives the rest of the outcome.

Payoffs and assumptions

Payoffs for each project π:

$$U_a(\pi) = \rho(\pi)\pi - \sigma(\pi) \qquad U_i(\pi) = \big(1 - \rho(\pi)\big)\pi - \big(I - \sigma(\pi)\big)$$

Budget constraints: $B_a \geq \int \sigma(\pi) f(\pi)\, d\pi$ and $B_i \geq \int (I - \sigma(\pi)) f(\pi)\, d\pi$

Assumptions

(A1) $\mathbb{E}(\pi) - I < 0 < 1 - I$, and f is log-concave. The average project is not worth financing, the best one is.

(A2) $B_a < I(1 - F(I)) \leq B_a + B_i$. The agency cannot finance all good projects alone, but the agency and the investors together can.

(A3) The agency is profit-maximizing.

Game 1 (Benchmark)

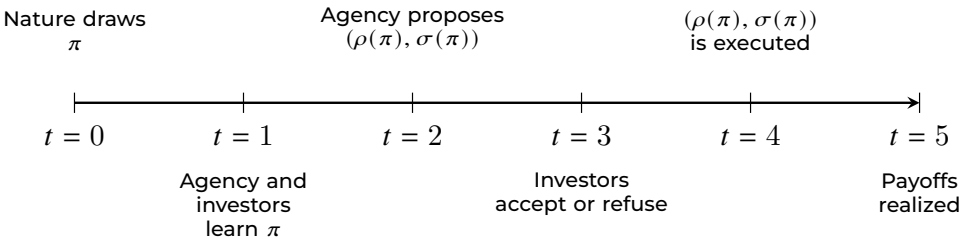


Figure: Game 1 – Benchmark

Game 1 (Benchmark)

Proposition 1

Under budget and investors participation constraints:

The agency maximizes profit	$\Leftrightarrow$	Investor utility $U_i(\pi) = 0$ for each π
		The agency finances all good projects

Investors participation constraint: $U_i(\pi) \geq 0$ for each π

Game 1 (Benchmark)

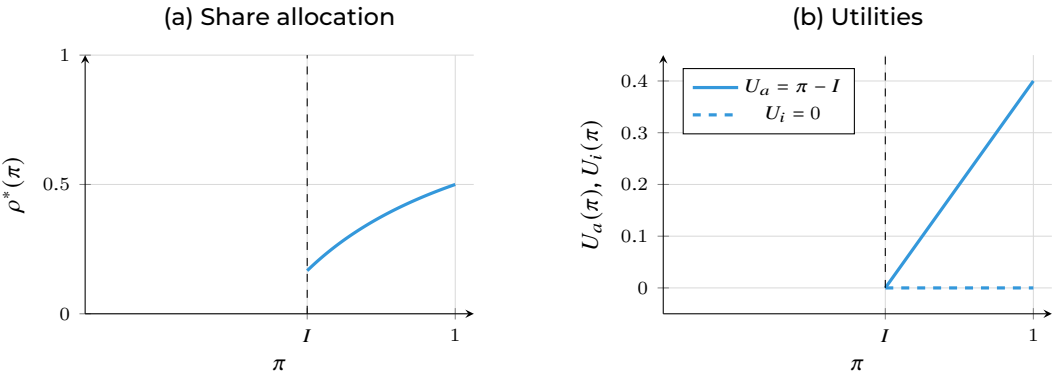


Figure: Parameters: $\pi \sim U[0,1]$, $I = 0.6$, $\sigma^* = 0.1$. (a) The share $\rho^*(\pi) = 1 - (I - \sigma^*)/\pi$ is increasing in π. (b) The agency captures the full surplus $\pi - I$; the investor earns zero rent.

Game 2 (Ex-post commitment)

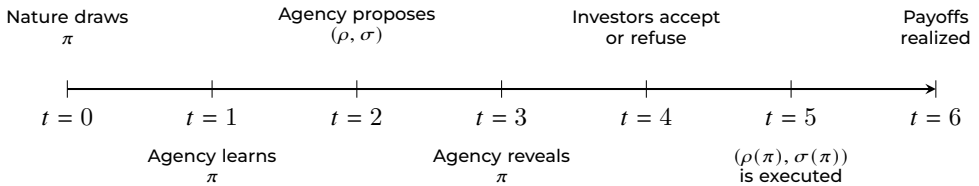


Figure: Game 2 – Ex-post commitment

Game 2 (Ex-post commitment)

Proposition 2

Under budget and investors participation constraints:

The agency maximizes profit	$\Leftrightarrow$	The allocation is pooling The agency saturates its budget $U_i(\pi_{\min}) = 0$ for the less good financed project The profit-maximization FOC is satisfied

Investors participation constraints:

- $U_i(\pi) \geq 0$ for each π
- $U_a(\pi) \geq \rho(\hat{\pi})\pi - \sigma(\hat{\pi})$ for each $(\pi, \hat{\pi})$

Game 2 (Ex-post commitment)

Remark

$\pi_{\min} > I$, so some good projects always remain unfunded.

A contract for each project

Four projects, $I = 1/2$, $B_a = 0.4$. Tailored terms, every investor exactly at zero.

Project	$\pi = \frac{1}{2}$	$\pi = \frac{1}{2}$	$\pi = \frac{3}{4}$	$\pi = 1$	
Contract (ρ, σ)	$(0.1,\ 0.05)$	$(0.1,\ 0.05)$	$(0.4,\ 0.05)$	$(0.6,\ 0.1)$	
Agency U_a	0	0	0.25	0.50	$\Sigma = 0.75$
Investor U_i	0	0	0	0	$\Sigma = 0.00$

The agency takes the whole surplus, 0.75, and spends 0.25 of its budget.

The agency is the one who says which project is which

It reports a $\pi = 1/2$ project as $\pi = 1$: $0.6 \cdot \frac{1}{2} - 0.1 = 0.20 > 0$.

reported as $\pi = 1$

Project	$\pi = \frac{1}{2}$	$\pi = \frac{1}{2}$	$\pi = \frac{3}{4}$	$\pi = 1$	
Contract (ρ, σ)	$(0.1,\ 0.05)$	$(0.6,\ 0.1)$	$(0.4,\ 0.05)$	$(0.6,\ 0.1)$	
Agency U_a	0	0.20	0.25	0.50	$\Sigma = 0.95$
Investor U_i	0	-0.20	0	0	$\Sigma = -0.20$

Every project below $\pi = 1$ wants that contract, and the constraint $U_i(\pi) \geq 0$ is written project by project.

One contract for all, and a saturated budget

The best single contract is $(0.6,\ 0.2)$: it funds $\pi \geq 3/4$ and spends $2 \times 0.2 = B_a$.

Project	$\pi = \frac{1}{2}$	$\pi = \frac{1}{2}$	$\pi = \frac{3}{4}$	$\pi = 1$	
Contract (ρ, σ)	not funded	not funded	$(0.6,\ 0.2)$	$(0.6,\ 0.2)$	
Agency U_a	—	—	0.25	0.40	$\Sigma = 0.65$
Investor U_i	—	—	0	0.10	$\Sigma = 0.10$

The contract pools, saturates the budget and sets $U_i(\pi_{\min}) = 0$, with $\pi_{\min} = 3/4 > I = 1/2$.

Game 2 (Ex-post commitment)

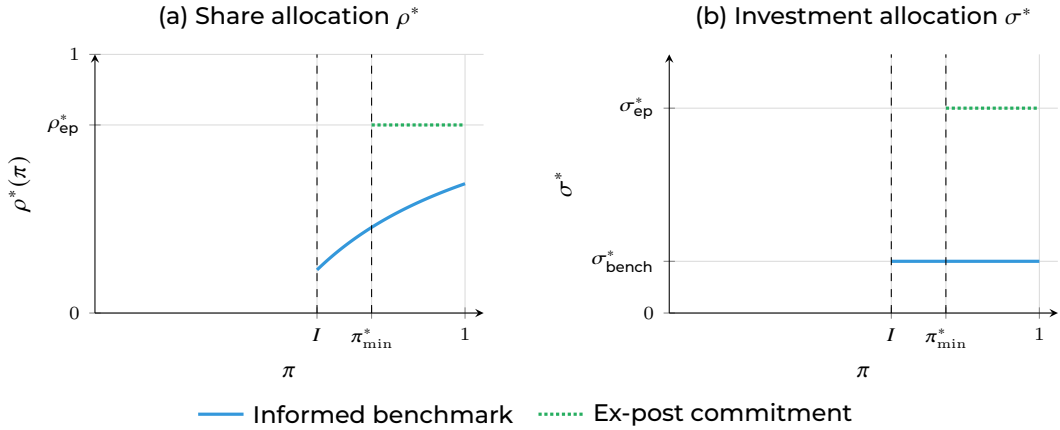


Figure: Informed benchmark vs. ex-post commitment regime. Parameters: $\pi \sim U[0,1]$, $I = 0.6$, $B_a = 0.1$, $B_i = 0.2$.

Game 2 (Ex-post commitment)

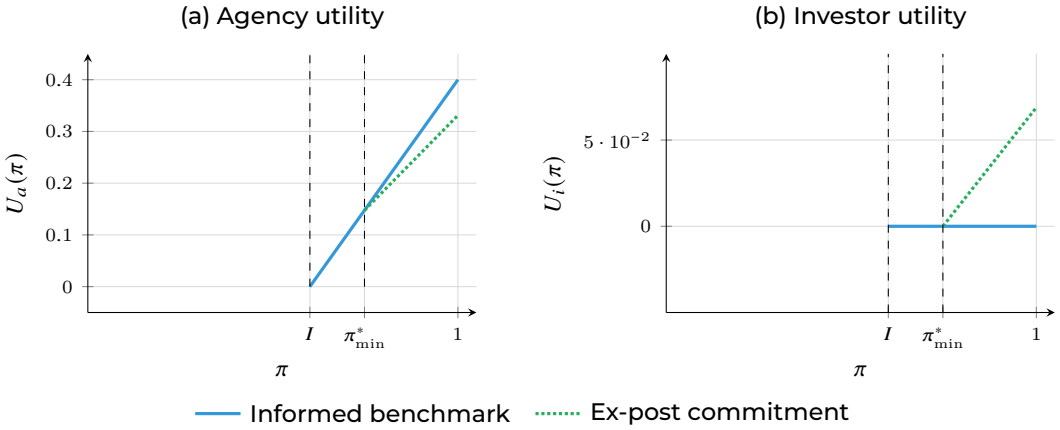


Figure: Utilities per project: informed benchmark vs. ex-post commitment regime. Same parameters.

Game 3 (Ex-ante commitment)

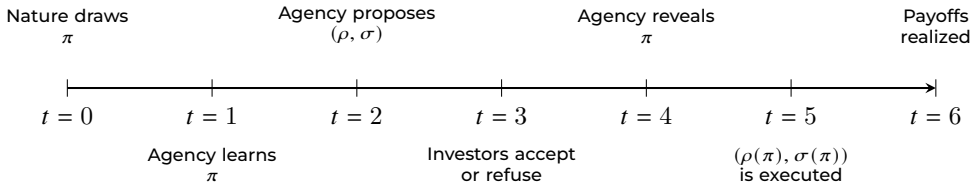


Figure: Game 3 – Ex-ante commitment

Game 3 (Ex-ante commitment)

Proposition 3

Under budget and investors participation constraints:

The agency maximizes profit $\Longleftarrow$

- The allocation is pooling
- The agency does not saturate its budget
- Investor average utility $\int U_i(\pi) f(\pi) d\pi = 0$
- The agency finances all good projects

Investors participation constraint: $\int U_i(\pi) f(\pi) d\pi \geq 0$

Game 3 (Ex-ante commitment)

Remark

The reverse is not true.

Game 3 (Ex-ante commitment)

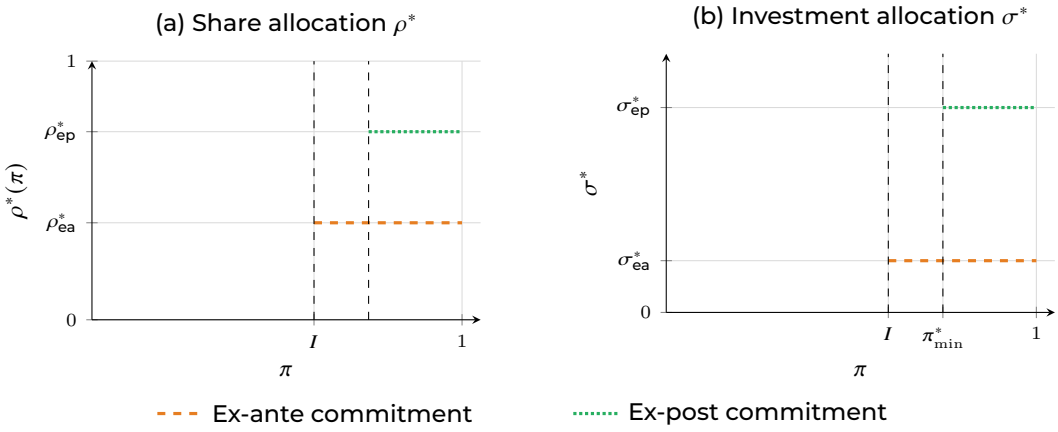


Figure: Optimal allocation under ex-ante vs. ex-post commitment investors. Same parameters as in Game 2.

Game 3 (Ex-ante commitment)

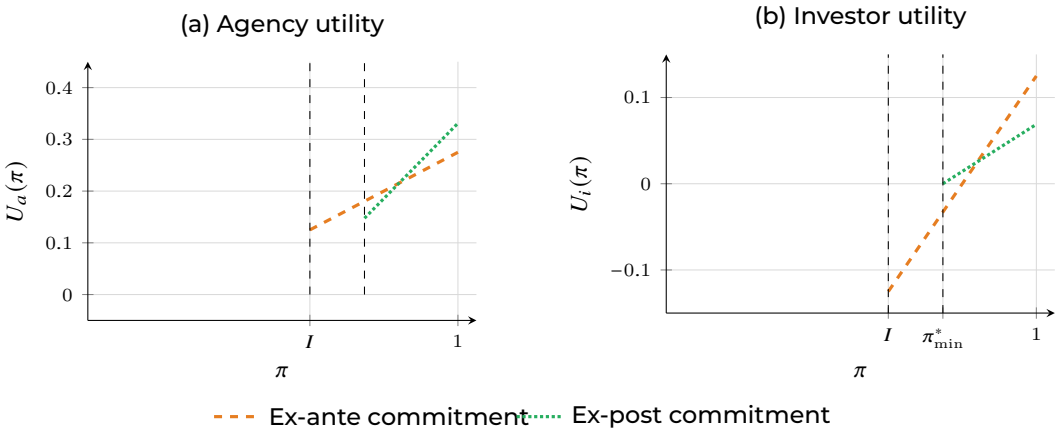
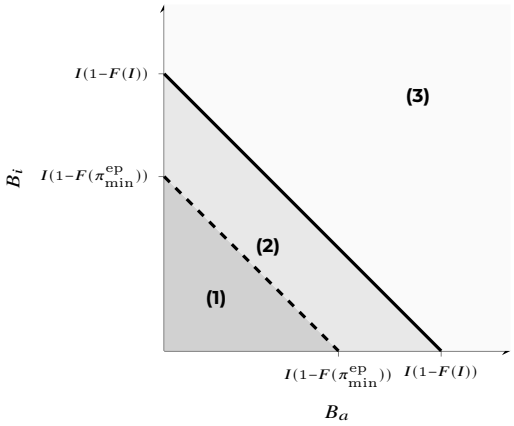


Figure: Utilities under the two commitment regimes. Same parameters as in Game 2.

Cash-poor vs. cash-rich economies

The friction is slack only where capital itself is short.



(1) Severely cash-poor: both regimes coincide
(2) Moderately cash-poor: ex-post excludes more
(3) Cash-rich: ex-post excludes $[I, \pi^{\text{ep}}_{\min})$

Where public money should enter

The state as a limited partner, not as an accelerator.

Financing threshold	Public accelerator ex-post commitment	Public LP ex-ante commitment
(3) Cash-rich	$\pi^{\text{ep}}_{\min} > I$	I
(2) Moderately cash-poor	$\pi^{\text{ep}}_{\min}$	$(1-F)^{-1}\left(\frac{B_a+B_i}{I}\right) < \pi^{\text{ep}}_{\min}$
(1) Severely cash-poor	$(1-F)^{-1}\left(\frac{B_a+B_i}{I}\right)$	$(1-F)^{-1}\left(\frac{B_a+B_i}{I}\right)$

EIF and Bpifrance already operate this way.

Takeaways

Main result

- Ex-ante: pooling is optimal, but not uniquely.
- Ex-post: pooling is unique, budget-saturating, inefficient.

- Pooling contracts are optimal.
- Accelerators as early-stage financiers induce an inefficiency.
- Public capital can efficiently intervene as LP.

Thank you

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